

index.qmd

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Introduction

This data was obtained from <https://www.waterpointdata.org/2021/11/09/global-water-challenge-announces-the-development-of-whdx-an-open-data-exchange-to-improve-wash-in-healthcare-facilities/>

under the aquaya waterTracs program.

private information such as ids were removed. in the dictionary file you will find important variables that may need some interpretation.

PROJECT DESCRIPTION

To identify the organisations that constructed the different water points

To assess their safety as drinking water points

To identify who manages the water points.

Methods

Involved renaming a few columns as well as mutating some to give specific values. The Na values could not be removed as they would make the results meaningless.

```
library(tidyverse)
library(here)
```

```
raw_data <- read_csv(here::here("data/raw/CAPSTONE.csv"))
```

```
raw_data_new <- raw_data |>
  rename(year = install_year,
         fecal_value = fecal_coliform_value,
         fecal_presence = fecal_coliform_presence,
         country_id = clean_country_id,
         country_name = clean_country_name,
         adm1 = clean_adm1,
         tech = water_tech_clean,
         management = management_clean,
         status = status_clean
  )
```

raw_new_data is the processed folder being used.

```
raw_data_new
```

```
# A tibble: 1,000 x 25
```

	row_id	source	report_date	status_id	water_source	water_tech	year
	<dbl>	<chr>	<dtm>	<chr>	<chr>	<chr>	<dbl>
1	9063	Water For~	2020-02-19 00:00:00	Yes	<NA>	Protected~	NA
2	9223	Water For~	2005-09-07 00:00:00	No	<NA>	Protected~	NA
3	9376	Water For~	2020-02-20 00:00:00	No	<NA>	Protected~	NA
4	13241	Water For~	2019-03-15 00:00:00	Yes	<NA>	Protected~	NA
5	15068	Water For~	2019-03-19 00:00:00	Yes	<NA>	Protected~	NA
6	19412	Water For~	2005-09-07 00:00:00	Yes	<NA>	Protected~	NA
7	31145	Water For~	2019-03-21 00:00:00	No	<NA>	Protected~	NA
8	32000	Water For~	2020-02-20 00:00:00	Yes	<NA>	Protected~	NA
9	39965	Water Mis~	2013-12-12 00:00:00	Unknown	<NA>	Electrica~	2012
10	41933	Water Mis~	2013-12-18 00:00:00	Unknown	<NA>	Electrica~	2008

```
# i 990 more rows
```

```
# i 18 more variables: installer <chr>, fecal_value <lgl>,
```

Table 1: Presence and absence counts in Uganda

country_id	fecal_presence	n
GTM	NA	1
HND	NA	10
KEN	NA	1
MEX	NA	4
UGA	Absent	1
UGA	Present	75
UGA	NA	908

```
# fecal_presence <chr>, activity_id <chr>, country_id <chr>,
# country_name <chr>, adm1 <chr>, water_source_clean <chr>,
# water_source_category <chr>, tech <chr>, `_facility_type` <chr>,
# management <chr>, status <chr>, count <dbl>, updated <dtm>,
# pay_clean <chr>, subjective_quality_clean <chr>, dataset_title <chr>
```

```
library(gt)
library(gapminder)
```

Results

Creating tables and figures or graphs showing the relationship between fecal contamination at water points and other factors such as the installer, the country, and the management system.

```
summary_table_country <- raw_data_new |>
  count(country_id, fecal_presence)

summary_table_country |>
  gt() |>
  fmt_number(columns = c(country_id, fecal_presence),
             decimals = 1)
```

?@tbl-country__fecal shows the number of water sources with presence of fecal contamination.

```
ggplot(summary_table_country, aes(x = country_id, y = n, fill = fecal_presence)) +
  geom_col(position = "stack")+
  labs(title = "Presence and absence counts in Uganda", x = "Country", y = "number" ) +
  theme_minimal()
```

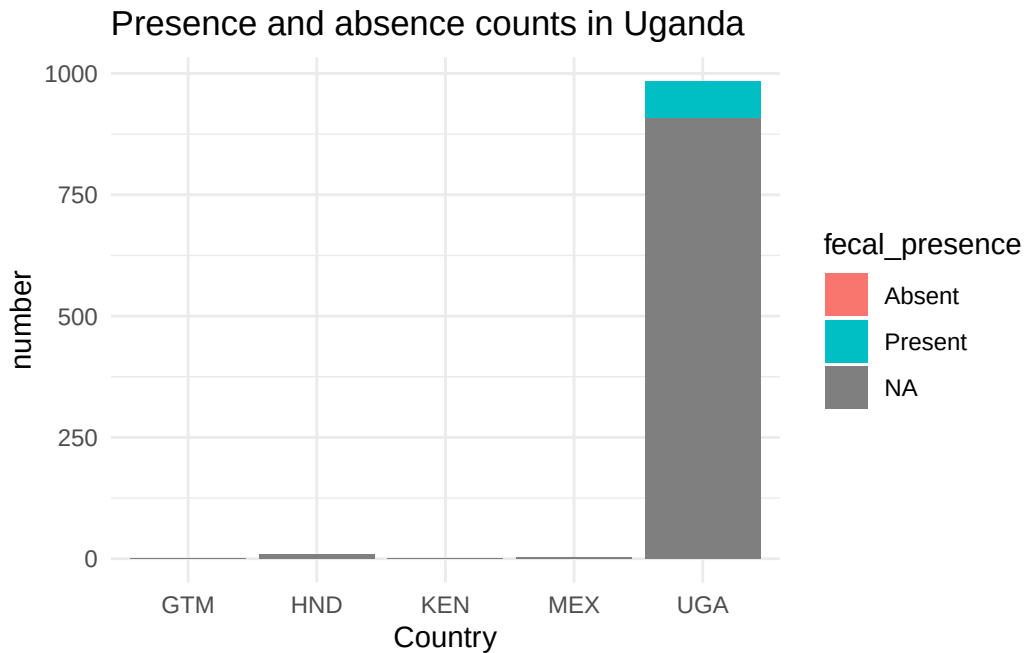


Figure 1: Graph showing presence and absence counts in Uganda

Figure 1 shows a graph with the presence and absence of fecal contamination at water sources.

```
raw_data_new <- raw_data_new |>
  mutate(management = case_when(
    management == "Community Management" ~ "CM",
    management == "Other" ~ "Ot",
    management == "Other Institutional Management" ~ "OIM",
    management == "Private Operator/ Delegated Management" ~ "PO/DM",
    management == "NA" ~ "NA"
  ))
```

```
summary_table_management <- raw_data_new |>
  count(management, fecal_presence)

summary_table_management |>
  gt() |>
  fmt_number(columns = c(management, fecal_presence),
    decimals = 1)
```

Table 2 shows the different management systems and the presence of fecal contamination.

Table 2: Presence and absence counts in management systems

management	fecal_presence	n
CM	NA	547
OIM	NA	60
Ot	NA	11
NA	Absent	1
NA	Present	75
NA	NA	306

```
ggplot(summary_table_management, aes(x = management, y = n, fill = fecal_presence)) +
  geom_col(position = "stack")+
  labs(title = "Presence and absence counts in different managements", x = "management system") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

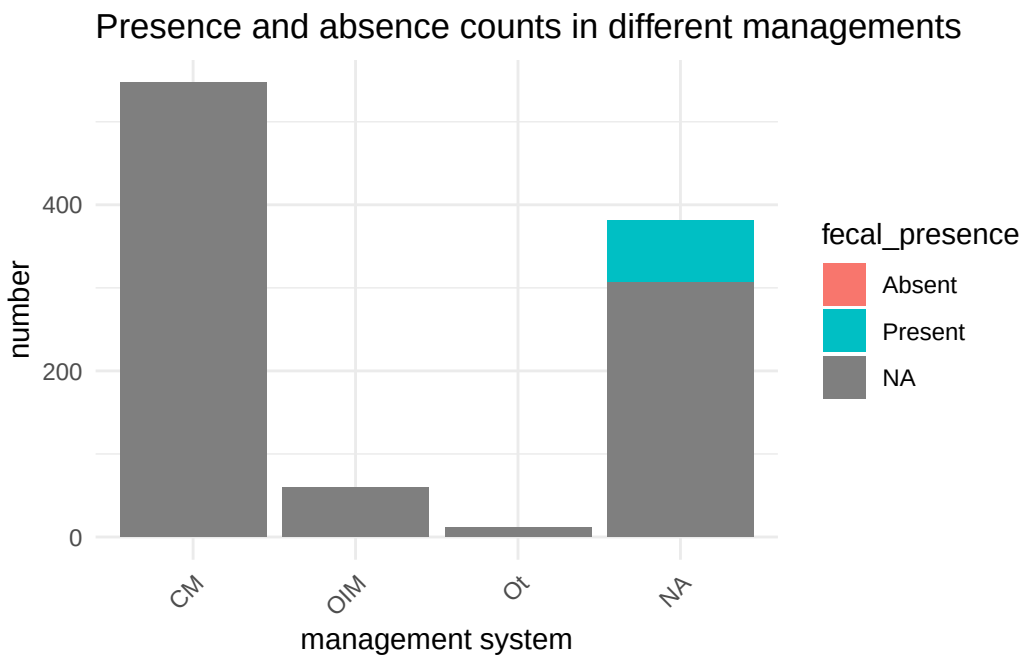


Figure 2: Graph showing presence and absence counts in management systems

Figure 2 shows a graph that depicts results in the table.

Table 3: Presence and absence counts in water sources by different installers

installer	fecal_presence	n
Water Mission	NA	18
World Vision	NA	13
other	NA	1
NA	Absent	1
NA	Present	75
NA	NA	892

```
summary_table_installer <- raw_data_new |>
  count(installer, fecal_presence)

summary_table_installer |>
  gt() |>
  fmt_number(columns = c(installer, fecal_presence),
             decimals = 1)
```

Table 3 shows the fecal presence where different organisations have installed water points.

```
ggplot(summary_table_installer, aes(x = installer, y = n, fill = fecal_presence)) +
  geom_col(position = "stack")+
  labs(title = "Presence and absence counts in water sources by different installers", x = "installer", y = "n") +
  theme_minimal()
```

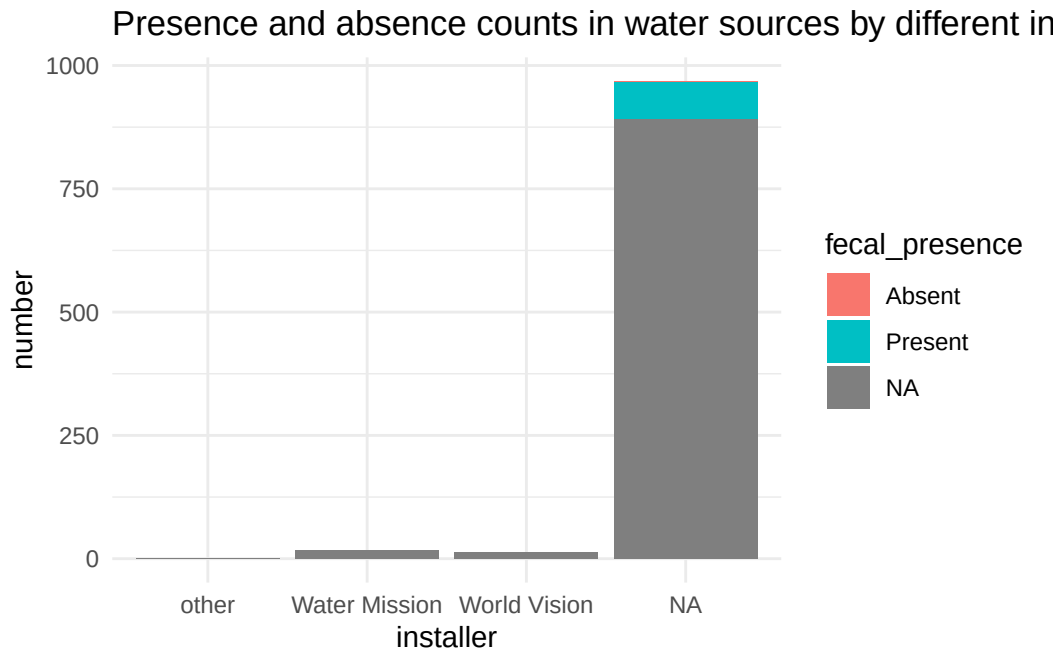


Figure 3: Graph showing presence and absence counts in water sources by different installers

Figure 3 shows fecal contamination in a graph format.

Conclusions

Fecal contamination in water sources is existent where communities are in charge of managing water sources.

It is also prominent where other organisations that have not been recognised are in charge of constructing water points.

Uganda has been seen to have the highest water sources with fecal contamination and others are unknown.

The study suggested that improving sanitary completion and local environmental hygiene was more important than controlling on-site sanitation in improving the quality of these springs.

References

[@howard2003; @pedley1997; @powell2003]